

2. System Architecture

Architecture Overview

The system follows a layered architecture with three main components:

1. Frontend (User Interface)
2. Backend API
3. Route Planning Module

These components communicate through a REST API.

Frontend

The frontend is responsible for user interaction and visualization.

Technologies:

- React
- Leaflet
- OpenStreetMap tiles

Responsibilities:

- display map
- allow user to select mission parameters
- visualize obstacles
- visualize route
- simulate drone movement

Backend

The backend manages system logic and communication with the route planner.

Technologies:

- FastAPI
- Python

Responsibilities:

- receive mission requests
- validate input data
- retrieve obstacle data
- call route planning algorithm
- return route results

Planning Module

The planning module generates the drone route.

Responsibilities:

- process obstacle geometry
- construct internal planning model
- compute route
- return optimized path

System Flow

User → Frontend (React) → Backend API (FastAPI)
→ Route Planner → Backend Response → Frontend Visualization